

## CS 171: Problem Set 4

Due Date: October 10th, 2026 at 8:59pm via Gradescope

### 1. The DES Complementation Property (10 points)

The DES block cipher is a 16-round Feistel network with a block size of 64 bits. Its key schedule derives a sequence of 48-bit round keys  $k_1, \dots, k_{16}$  from the master key. The details of the key schedule are not important for this exercise.

The DES round function  $\widehat{f}$  takes a 48-bit round key  $k_i \in \{0, 1\}^{48}$  and a 32-bit input  $R \in \{0, 1\}^{32}$ . It is evaluated as follows:

1. The expansion function  $E$  expands  $R$  to 48 bits by duplicating certain bits of  $R$ .
2. The value  $E(R)$  is XORed with  $k_i$ .
3. The resulting 48-bit string is divided into eight 6-bit blocks. Each block is passed through an S-box that maps 6 bits to 4 bits.
4. A fixed permutation is applied to the resulting 32-bit string.

For a bit string  $y$ , let  $\overline{y}$  denote its bitwise complement. Prove that, for every sequence of round keys  $k_1, \dots, k_{16}$  and every input  $x \in \{0, 1\}^{64}$ ,

$$\text{DES}_{k_1, \dots, k_{16}}(x) = \overline{\text{DES}_{\overline{k_1}, \dots, \overline{k_{16}}}(\overline{x})}.$$

## 2. Insecure Candidates for MACs (10 points)

Two candidate MAC constructions are given below. Both use a pseudorandom function  $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ . Show that each construction is insecure.

### 1. Scheme 1:

(a)  $\text{Gen}(1^n)$ : Sample  $k \xleftarrow{\$} \{0, 1\}^n$  and output  $k$ .

(b)  $\text{Mac}(k, m)$ : Parse  $m = m_0 \parallel m_1$ , where  $m_0, m_1 \in \{0, 1\}^n$ , and output

$$t = F(k, m_0) \parallel F(k, m_0 \oplus m_1).$$

(c)  $\text{Verify}(k, m, t)$ : Output 1 if  $t = \text{Mac}(k, m)$ , and output 0 otherwise.

### 2. Scheme 2:

(a)  $\text{Gen}(1^n)$ : Sample  $k \xleftarrow{\$} \{0, 1\}^n$  and output  $k$ .

(b)  $\text{Mac}(k, m)$ : Parse  $m = m_0 \parallel m_1$ , where  $m_0, m_1 \in \{0, 1\}^{n-1}$ . Sample  $r \xleftarrow{\$} \{0, 1\}^n$  and output

$$t = r \parallel (F(k, r) \oplus F(k, 0 \parallel m_0) \oplus F(k, 1 \parallel m_1)).$$

(c)  $\text{Verify}(k, m, t)$ : Parse  $m = m_0 \parallel m_1$ , where  $m_0, m_1 \in \{0, 1\}^{n-1}$ , and parse  $t = r \parallel t'$ , where  $r, t' \in \{0, 1\}^n$ . Output 1 if

$$t' = F(k, r) \oplus F(k, 0 \parallel m_0) \oplus F(k, 1 \parallel m_1),$$

and output 0 otherwise.

### 3. Comparing Numbers Privately (Optional)<sup>1</sup>

Let  $p$  be prime. Alice holds  $a \in \mathbb{Z}_p$ , and Bob holds  $b \in \mathbb{Z}_p$ . They want Alice to learn whether  $a = b$  without revealing their numbers to a server, Sam. Assume that Sam follows the protocol, does not collude with either party, and has a private channel to each of them. All arithmetic below is in  $\mathbb{Z}_p$ .

For one equality test, Alice and Bob share independent, uniformly random keys  $k_0, k_1 \in \mathbb{Z}_p$ . Alice sends

$$x_a = a + k_0$$

to Sam. Bob chooses  $r$  uniformly from  $\mathbb{Z}_p$  and sends Sam a pair  $(r, x_b)$ . Sam computes

$$x = rx_a - x_b$$

and sends  $x$  to Alice. Alice concludes that  $a = b$  if  $x + k_1 = 0$ .

1. Determine how Bob should compute  $x_b$  from  $b, r, k_0, k_1$ . Show that Alice's test is always correct when  $a = b$ , and find its error probability when  $a \neq b$ . Explain why Sam's view reveals nothing about  $a$  or  $b$ . Explain why, when  $a \neq b$ , the value  $x + k_1$  reveals nothing about  $b$  to Alice.
2. Suppose Alice and Bob use the same pair  $(k_0, k_1)$  for two tests: one on inputs  $(a, b)$  and another on inputs  $(a', b')$ . Bob chooses fresh, independent values  $r$  and  $r'$  for the two tests. Show what goes wrong with privacy when the keys are reused.
3. Part 2 shows that Alice and Bob need a fresh key pair for every test. Rather than sharing many independent key pairs in advance, let  $B = B(n)$  be a polynomial and suppose that they share a single long-term key  $K \in \{0, 1\}^n$  for a secure pseudorandom function

$$F : \{0, 1\}^n \times \{0, 1\}^n \longrightarrow \mathbb{Z}_p^2.$$

Design a protocol for up to  $B(n)$  equality tests, and briefly justify why the resulting tests have the privacy properties of the one-time protocol.

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<sup>1</sup>This optional exercise is adapted from Dan Boneh's *A Graduate Course in Cryptography*.